

Sensor-Based Attendance Management and Activity Tracking system using Wi-Fi Module

Introduction

In the modern era of automation and smart systems, managing attendance and monitoring human activities efficiently have become essential in educational institutions, offices, and industrial environments. Traditional attendance systems that rely on manual entry or biometric scanning are often time-consuming, prone to errors, and require human supervision. To overcome these limitations, a sensor-based attendance management and activity tracking system integrated with a WiFi module provides an intelligent, real-time, and contactless solution. This system utilizes sensors such as PIR (Passive Infrared), ultrasonic, or RFID to detect presence, identity, and movement, while a microcontroller (like ESP32 or ESP8266) processes the collected data. The WiFi module enables seamless wireless communication between the sensing unit and a central server or cloud database, where attendance and activity records are automatically updated. Administrators can monitor data through a web or mobile interface, ensuring transparency and convenience. This technology not only reduces human effort and administrative workload but also enhances accuracy, efficiency, and reliability in attendance tracking and activity analysis. Hence, the integration of sensors with IoT and WiFi technology creates a smart, cost-effective, and scalable solution for modern attendance management systems.

Literature Review

In the work titled "Real-Time Smart Attendance System Using Face Recognition Techniques," Sawhney et al. proposed a solution to address the challenges faced by teachers when managing attendance manually [1]. Traditional attendance management can be burdensome, leading to the adoption of smart and automated systems. However, authentication poses a significant problem in such systems. Biometric techniques are commonly employed in smart attendance systems, with face recognition being one of them. Facial recognition, a crucial component of biometric authentication, finds applications in various domains such as network security, video surveillance, CCTV systems, human-computer interfaces, and indoor access control systems. By incorporating face recognition, this architecture offers a solution to combat issues like proxy attendance and false marking of students' presence when they are not physically present. In the following work [8], the focus is on attendance monitoring utilizing Radio Frequency Identification (RFID) technology. The paper proposes the integration of ubiquitous computing systems into classrooms to manage student attendance efficiently. RFID technology is recognized as a valuable tool for monitoring attendance and enhancing classroom security. By employing RFID technology, it becomes possible to automatically record the movements and locations of students within a classroom or school/university environment. The paper presents the implementation of a real-time intelligent system in conjunction with RFID hardware to capture students' attendance during lectures and

laboratory sessions in a school/university setting. Notably, this approach does not incur additional costs beyond the initial investment and offers a long-lasting solution. There are few other kinds of literature, discussing image

enhancements in low-light situations [3]. Poor visibility not only has implications for consumer photography but also affects computer vision applications. Low-light image enhancement is crucial for improving the visibility and perception of information in images for human viewers and image understanding. By manipulating the pixel values in digital images captured under low illumination, the details of low-light images can be enhanced. The primary objective of enhancing low-vision images is to increase their visibility.

Privacy-Preserving Zero-effort Class Attendance Tracking proposes a system with student smartphones in the classroom and produces "location proofs" backed up by the radio frequency fingerprints those devices scan, which are then used to confirm the students' positions [4]. The report suggests using instructor and student smartphones along with Wi-Fi access points installed in campus buildings to establish an effortless and private attendance tracking system. The system is unique because it doesn't necessitate any effort from users. Additionally, it protects user privacy because neither user identities nor class locations are stored on the App server.

In situations where the Wi-Fi network experiences slow connectivity, our smart attendance system adapts by providing an alternative solution. By utilizing the mobile data on their phones and accessing the GPS functionality, staff members can conveniently mark their attendance [5].

Research Gap

Although several studies have explored smart and automated attendance systems using technologies such as face recognition, RFID, and IoT, existing research still presents notable limitations that create opportunities for further development. Sawhney et al. (2019) proposed a face recognition-based attendance system integrated with cloud computing, which achieved automation but relied heavily on high-quality imaging and stable lighting conditions. This dependence limits performance in low-light or outdoor environments. Similarly, Nasrin Jumana et al. (2023) addressed image enhancement for low-light scenarios, but their work focused primarily on image preprocessing rather than end-to-end attendance management or activity tracking.

Shrivastava et al. (2023) designed an IoT-based RFID attendance monitoring system using Arduino and ESP8266; however, their system only supports identity-based attendance without incorporating behavioral or activity analysis, and it lacks real-time environmental adaptability or motion-based detection. Shene et al. (2021) introduced a privacy-preserving, zero-effort attendance system, which effectively minimized user intervention but did not utilize low-cost sensor-based modules suitable for scalable deployment in classrooms or offices. Meanwhile, Zahid et al. (2023) developed a multi-stage CNN-based object and face detection approach, enhancing

recognition accuracy, but the method is computationally intensive, requiring powerful hardware unsuitable for small embedded or IoT-based systems.

Hence, the existing research reveals a gap in developing a low-cost, sensor-driven, WiFi-enabled system capable of performing automatic attendance detection and real-time activity tracking without relying on camera-based recognition. Current models emphasize either identification accuracy (via vision/RFID) or data transmission (via IoT), but few integrate presence sensing, activity monitoring, and wireless data synchronization into one framework. This gap highlights the need for an IoT and sensor-based attendance management system that ensures contactless operation, real-time data logging, energy efficiency, and privacy preservation—objectives that this study aims to fulfill through the integration of sensors and WiFi modules in a unified, intelligent system.

Rationale of the Study

In today's digital era, automation and smart technologies have become vital in improving efficiency, accuracy, and productivity in both academic and professional environments. Traditional attendance management methods—such as manual signing, biometric scanning, or card swiping—are often time-consuming, error-prone, and require continuous human supervision. These systems also face challenges related to data reliability, maintenance costs, and limited scalability. Hence, there is a growing need for an intelligent, contactless, and automated attendance management solution that minimizes human intervention while ensuring data accuracy and accessibility.

Furthermore, this system promotes efficiency, transparency, and scalability in various environments such as classrooms, offices, and industries. It supports contactless operation, which is especially beneficial in post-pandemic scenarios where health and hygiene are critical. The inclusion of activity tracking helps analyze behavior patterns, space utilization, and work engagement, offering valuable insights for organizational management. Overall, the rationale for this study is to develop a smart, reliable, and low-cost IoT-based solution that revolutionizes conventional attendance systems by combining sensor technology, WiFi communication, and intelligent data processing to create a fully automated and real-time attendance and activity tracking framework.

Research Questions

In order to investigate the opportunities and difficulties of utilizing IoT and AI technologies to manage attendance system and Tracking activities, this study tackles a number of research problems. The following are the research questions:

- a. What kinds of methods and IoT devices are being utilized to monitor and identify presence and employees activities?
- b. How can activity tracking be incorporated into the attendance system to analyze user presence?
- c. What are the primary obstacles and constraints to using IOT in wifi modules for managing Attendance System and activity tracking?

Objective

- 1.To design a reliable and automated attendance management system using sensors.
- 2.To track individual activities (entry, exit, and location movement) in real-time.
- 3.To eliminate manual interventions and reduce proxy or fraudulent attendance.
- 4.To store attendance and activity logs in a centralized database for future analysis.
- 5.To provide easy reporting and monitoring through a web or mobile dashboard.
- 6.To provide real-time monitoring of user activity within a campus,office or lab.
- 7.To generate reports for administration.
- 8.To improve security by monitoring movement inside restricted zones.

System Components

Biometric Sensor (Fingerprint/Face recognition): For unique identification.

RFID / NFC Tags: Alternative for quick identification.

PIR Motion Sensors: To track physical presence/activity in classrooms/offices.

Wi-Fi/Bluetooth Module: For wireless data transmission.

Cloud Database: To store attendance and activity records.

Web/Mobile Application: For monitoring, reporting, and analytics.

Methodology

The proposed Sensor-Based Attendance Management and Activity Tracking System using a WiFi Module follows a structured methodology involving hardware design, software development, data communication, and performance evaluation. The system begins with the integration of a microcontroller (such as ESP32 or ESP8266) that serves as the central processing and communication unit. Various sensors—including PIR (Passive Infrared) sensors, ultrasonic sensors, RFID/NFC readers, or accelerometers—are connected to the microcontroller to detect human presence, identity, and movement. When a person enters the monitored area, the sensors generate signals that are processed by the microcontroller to determine attendance status and activity type. Each detected event (such as presence, motion, or identification through RFID tag) is recorded and transmitted through the WiFi module to a central database or cloud server using MQTT or HTTP communication protocols.

To ensure system reliability, a debouncing algorithm is implemented to minimize false detections caused by sensor noise or environmental changes. The communication process is secured using encrypted data transmission to protect privacy and prevent unauthorized access. The methodology also includes calibration of sensors to optimize sensitivity and range, ensuring accurate detection within the intended coverage area. Testing is conducted under different environmental conditions to evaluate accuracy, latency, and connectivity stability. Performance metrics such as detection accuracy, data transmission delay, and false trigger rate are measured to assess system efficiency.

Overall, this methodology combines IoT technology, wireless communication, and sensor-based automation to create a reliable, low-cost, and real-time attendance and activity tracking system. It reduces manual errors, enhances productivity monitoring, and provides an intelligent framework suitable for classrooms, offices, or industrial environments.

Applications

Educational Institutions (student attendance tracking).

Corporate Offices (employee monitoring).

Research Labs and Libraries.

Secure Facilities requiring restricted access.

Reference

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